Fatemeh Rahimi

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Lead Data Scientist | NLP, LLM Systems, Applied AI

Lead Data Scientist with 7+ years of experience building production NLP systems, LLM workflows, and applied ML products. Strong background in document understanding, evaluation, retrieval, and practical AI systems that work under real-world constraints.

EXPERIENCE

Lead Data Scientist
Apr 2026 – Present

Pythonic AI | Remote

- Leading applied AI and data science work across production NLP systems and LLM-based workflows at Pythonic AI.
- Setting technical direction on evaluation strategy, model quality, and practical AI system design for production use cases.
- Supporting cross-functional delivery and mentoring teammates across research, annotation, and production-facing AI work.

Senior NLP & AI Scientist
Feb 2022 – Mar 2026

Pythonic AI | Remote

- Led development and deployment of 3 production NLP services for title and escrow document workflows, supporting extraction and classification across insurance, loan, mortgage, and signed closing documents.
- Served as technical owner for a production NLP service and drove model quality improvements through evaluation pipelines, release validation, and iteration on real-world failure modes.
- Architected an LLM-powered multi-agent workflow with tool orchestration and human-in-the-loop escalation to automate support email triage and resolution.
- Built internal automation tooling to streamline labeling, sprint planning, and QA workflows, and mentored teammates on NLP, evaluation, and applied AI best practices.

Applied Research Scientist, NLP
Nov 2020 – Jan 2022

Imagia | Montreal, Canada (Remote)

- Built NLP pipelines for clinical report analysis, including de-identification, named entity recognition, and measurement extraction.
- Applied and evaluated transformer-based models including BioBERT and ClinicalBERT on real-world biomedical datasets.
- Designed production-grade CLI tools for end-to-end inference on clinical documents.

NLP Research Assistant
Mar 2019 – Apr 2021

Dalhousie University | Halifax, Canada

- Applied and evaluated deep language models including BERT, BioBERT, and BlueBERT on downstream NLP tasks with rigorous experimental design.
- Developed MTLV, a multi-task learning library with 4 architectures and a novel task-clustering method, published at ACM DocEng 2021.

SELECTED PROJECTS

Multi-agent Support Automation: Designed an agentic support workflow combining tool-using LLM agents, retrieval, and human review to reduce manual engineering effort and speed up issue resolution.

Production NLP Services: Built and maintained document understanding systems for high-value business workflows, with emphasis on evaluation, reliability, and deployability.

SKILLS

Core: Python, SQL, PyTorch, scikit-learn, Transformers, spaCy
LLM Systems: RAG, LLM evaluation, Agentic workflows, Human-in-the-loop systems, Prompt engineering
Applied NLP: Information extraction, Text classification, Named entity recognition, Document understanding, Search and retrieval
Infra & Tooling: Docker, REST APIs, GitHub Actions, MLflow, CLI tooling

EDUCATION

M.Sc., Computer Science
Dalhousie University, Halifax, Canada
2019 – 2021 | GPA 4.07/4.3

B.Sc., Software Engineering
Shiraz University, Shiraz, Iran
2013 – 2018

PUBLICATION

MTLV: A Library for Building Multi-task Learning Architectures for NLP
ACM Symposium on Document Engineering (DocEng), 2021